

## 0 math voodoo

$$\ln x \leq x - 1.$$

$$e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \dots, \quad x \in \mathbb{R}.$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots, \quad |x| < 1.$$

Markov's inequality: For nonnegative  $X$ ,

$$\mathbb{P}(X \geq a) \leq \frac{\mathbb{E}[X]}{a}, \quad a > 0.$$

Chebyshev's inequality:

$$\mathbb{P}(|X - \mathbb{E}[X]| \geq a) \leq \frac{\text{Var}(X)}{a^2}, \quad a > 0.$$

Jensen's inequality: For convex  $f$ ,

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)].$$

## 1 Measures

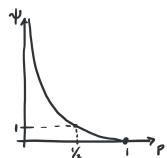
### 1.1 Information

To quantify the **amount of information** in an event occurring with probability  $p$ , we find a function  $\psi$  such that:

- $\psi(p)$  is **non-negative**.
- $\psi(1) = 0$ . Definite events carry no information.
- $\psi(p)$  is **continuous**, and decreasing in  $p$ . Unlikely events carry more information.
- $\psi(p_1 p_2) = \psi(p_1) + \psi(p_2)$ . Independent events carry additive information.

The only function that satisfies these axioms is, for some base  $b$ :

$$\psi(p) = \log_b \frac{1}{p} \quad p \in [0, 1].$$



We consider  $b = 2$  to measure information in **bits**. Any other base is equivalent up to a constant factor.

### 1.2 Shannon Entropy

Consider a discrete random variable  $X$  with PMF  $p(x)$ . To quantify the **entropy** of (the distribution of)  $X$  such that:

- $H(\mathbf{p})$  is **continuous** as a function of  $\mathbf{p}$ .
- $H(\mathbf{p})$  is **maximised for the uniform distribution**, and is increasing in the number of outcomes.
- Considering successive decisions, the following holds:

$$H(\mathbf{p}) = H(p_1 + p_2, \dots, p_N) + (p_1 + p_2) H_2\left(\frac{p_1}{p_1 + p_2}, \frac{p_2}{p_1 + p_2}\right)$$

The only function that satisfies these axioms is the **Shannon entropy**, up to a constant factor:

$$H(P_X) = \mathbb{E}_{X \sim P_X} \left[ \log_2 \frac{1}{P_X(X)} \right].$$

Entropy is a measure of **uncertainty/information** in  $X$ .

**Binary entropy**: In particular, for a Bernoulli distribution with probability  $p$ , the entropy is given by the **binary entropy function**:

$$H_2(p) = -p \log_2 p - (1-p) \log_2 (1-p).$$

**Uniform distribution**: For a uniform distribution over  $N$  outcomes, the entropy is given by:

$$H(\mathbf{p}) = \log_2 N.$$

Entropy generalises to **joint** and **conditional distributions**:

$$H(X, Y) = \mathbb{E}_{(X, Y) \sim P_{X, Y}} \left[ \log_2 \frac{1}{P_{X, Y}(X, Y)} \right],$$

$$H(X|Y) = \mathbb{E}_{(X, Y) \sim P_{X, Y}} \left[ \log_2 \frac{1}{P_{X|Y}(X|Y)} \right].$$

$H(X|Y)$  can be interpreted as the **uncertainty in  $X$  after observing  $Y$** .

### Properties of Shannon entropy

- $H(X) \geq 0$ . Entropy is **non-negative**, and is zero iff  $X$  is deterministic.
- $H(X) \leq \log_2 |\mathcal{X}|$ , with equality iff  $X$  is uniform.
- Chain rule**:

$$H(X_1, \dots, X_n) = \sum_{i=1}^n H(X_i | X_1, \dots, X_{i-1})$$

- $H(X|Y) \leq H(X)$ , with equality iff  $X, Y$  are independent. **NOTE**:  $H(X|Y = y)$  could exceed  $H(X)$ .
- $H(X_1, \dots, X_n) \leq \sum_{i=1}^n H(X_i)$ , with equality iff  $X_1, \dots, X_n$  are independent.
- Data processing inequality**: For any deterministic function  $f$ ,  $H(X) \geq H(f(X))$ .
- Information-preserving transformations**: For any bijection  $f$ , or if  $Y$  depends on  $X$  only through  $f(X)$ ,  $H(X) = H(f(X))$ .

- Concavity**:  $H(\lambda P + (1-\lambda)Q) \geq \lambda H(P) + (1-\lambda)H(Q)$

### 1.3 KL Divergence

To quantify the **difference between two distributions  $P$  and  $Q$** , we use the **KL Divergence**:

$$D_{\text{KL}}(P||Q) = \mathbb{E}_{X \sim P} \left[ \log_2 \frac{P(X)}{Q(X)} \right].$$

- $D(P||Q)$  is **non-negative**, and is zero iff  $P = Q$ .
- Chain Rule**: Let  $C := D(P_{Y|X} || Q_{Y|X} | P_X)$ .  $D(P_{XY} || Q_{XY}) = D(P_X || Q_X) + C$ ,  $C = \sum_x P_X(x) D(P_{Y|X}(\cdot|x) || Q_{Y|X}(\cdot|x))$
- Data processing inequality**: For any function  $f$ ,  $D(P_X || Q_X) \geq D(P_{f(X)} || Q_{f(X)})$ .

### 1.4 Mutual Information

To quantify the **"information gain"** of observing another variable, we use the **mutual information**:

$$I(X; Y) = H(X) - H(X|Y) = H(Y) - H(Y|X).$$

$$I(X; Y|Z) = H(X|Z) - H(X|Y, Z) = H(Y|Z) - H(Y|X, Z)$$

and **joint distributions**:

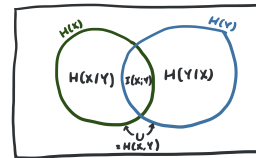
$$I(X; Y, Z) = H(X) - H(X|Y, Z).$$

Alternative forms:

$$I(X; Y) = D(P_{XY} || P_X P_Y)$$

$$= \mathbb{E} \left[ \log_2 \frac{P_{XY}(X, Y)}{P_X(X) P_Y(Y)} \right]$$

$$= \mathbb{E} \left[ \log_2 \frac{P_{Y|X}(Y|X)}{P_Y(Y)} \right]$$



### Properties of mutual information

- Non-negativity**:  $I(X; Y) \geq 0$ , with equality iff  $X$  and  $Y$  are independent.
- $I(X; Y) \leq \min \{H(X), H(Y)\}$  are **upper bounds** on information gain.
- $I(X; Y) = H(X)$  iff  $X$  is a **deterministic** function of  $Y$ .
- Symmetry**:  $I(X; Y) = I(Y; X)$ .
- Chain rule**:

$$I(X_1, \dots, X_n; Y) = \sum_{i=1}^n I(X_i; Y | X_1, \dots, X_{i-1})$$

- Data processing inequality**: If  $X \rightarrow Y \rightarrow Z$  forms a Markov chain, then

$$I(X; Z) \leq I(X; Y).$$

- Partial sub-additivity**: If  $Y_1, \dots, Y_n$  are conditionally independent given  $X_1, \dots, X_n$ , and  $Y_i$  depends on  $X_1, \dots, X_n$  only through  $X_i$ , then

$$I(X_1, \dots, X_n; Y_1, \dots, Y_n) \leq \sum_{i=1}^n I(X_i; Y_i).$$

- Convexity**:  $I(X; Y)$  is convex in  $P_{Y|X}$  for fixed  $P_X$ , and concave in  $P_X$  for fixed  $P_{Y|X}$ .

## 2 Symbol-wise Source Coding

### Setup

We denote the set of all symbols (the **alphabet**) by  $\mathcal{X}$ . Consider a discrete (memoryless) source  $X$  with PMF  $P_X$  over  $\mathcal{X}$ .

**Lowercase English text**:  $\mathcal{X} = \{a, \dots, z\}$ , with  $P_X(e)$  being the highest.

We encode each string  $x \in \mathcal{X}^*$  into a decodeable binary string  $C(x)$ , of length  $\ell(x)$ . The goal is to minimize the **expected codeword length** of the code  $C$ :

$$L(C) = \mathbb{E}_{X \sim P_X} [\ell(X)] = \sum_{x \in \mathcal{X}} P_X(x) \cdot \ell(x),$$

### 2.1 Symbol-Wise Coding

**Symbol-wise coding** encodes **each symbol**  $x \in \mathcal{X}$  into a codeword  $C(x)$ , and **encodes a string**  $x_1 \dots x_n$  by concatenating the codewords  $C(x_1) \dots C(x_n)$ .

### Decodeability Conditions

**Non-singular codes**: A code is non-singular if all codewords are distinct.

$$\forall x, x' \in \mathcal{X}, x \neq x' \implies C(x) \neq C(x').$$

**Uniquely decodable codes**: A code is uniquely decodable if any concatenation of codewords can be uniquely parsed into codewords.

That is, for distinct  $x_1 \dots x_n, x'_1 \dots x'_m \in \mathcal{X}^*$ ,  $C(x_1) \dots C(x_n) \neq C(x'_1) \dots C(x'_m)$ .

Uniquely decodeability is **non-trivial to verify**.

**Prefix-free codes**: A code is prefix-free if no codeword is a prefix of another codeword.

$C$  is prefix-free  $\implies C$  is uniquely decodable.

Additionally, prefix-free codes can be decoded **instantaneously**, without needing to wait for the entire string to be received.

## Kraft's inequality

Kraft's inequality states that for any prefix-free code  $C$ ,

$$\sum_{x \in \mathcal{X}} 2^{-\ell(x)} \leq 1.$$

Any prefix-free code can be **represented as a binary tree**, where each codeword corresponds to a leaf. The depth of the leaf is the codeword length. It follows, by random traversal of the tree, that the **probability of reaching a leaf** is  $2^{-\ell(x)}$ .

Conversely, for any set of codeword lengths  $\{\ell(x)\}_{x \in \mathcal{X}}$  satisfying Kraft's inequality, there exists a prefix-free code with these codeword lengths.

### Entropy lower bound

Entropy provides a fundamental **limit on the expected codeword length** of any uniquely decodable code:

$$L(C) \geq H(X),$$

with equality iff  $P_X(x) = 2^{-\ell(x)}$  for all  $x \in \mathcal{X}$ .

Using the entropy lower bound, **Shannon-Fano coding** is a simple method to construct a **prefix-free code for a given distribution  $P_X$** .

Ordering the symbols in decreasing probability, define:

$$C(x) = \text{binary expansion of } \sum_{x' < x} P_X(x'),$$

truncated to  $\ell(x) = \lceil -\log_2 P_X(x) \rceil$  bits.

A Shannon-Fano coding satisfies the **expected codeword length** of  $H(X) \leq L(C_{SF}) \leq H(X) + 1$ .

**Mismatched coding**: If the true distribution is  $P_X$  but **lengths are chosen according to  $Q_X$** , the Shannon-Fano code has expected codeword length

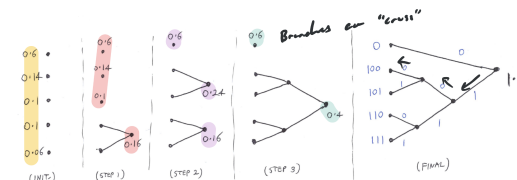
$$L(C_Q) = H(P_X) + D(P_X || Q_X).$$

### 2.2 Huffman Coding

Huffman coding constructs an **optimal prefix-free code** for a given distribution  $P_X$ . The **algorithm** is as follows:

- Start with a forest of  $|\mathcal{X}|$  trees, each consisting of a single node corresponding to a symbol  $x \in \mathcal{X}$ , with weight  $P_X(x)$ .
- While there are more than one tree in the forest:
  - Select the two roots with the smallest weights, say  $T_1$  and  $T_2$ .
  - Merge  $T_1$  and  $T_2$  into a new tree  $T$ , where  $T_1$  and  $T_2$  are the left and right subtrees of  $T$ , respectively. The weight of  $T$  is the sum of the weights of  $T_1$  and  $T_2$ .
- The remaining tree in the forest is the Huffman tree.

At each node, assign a binary digit to each branch. The code for a given symbol is the **sequence of binary digits along the path** from the root to the leaf corresponding to that symbol.



**Optimality**: The expected codeword length of the Huffman code is

$$L(C_H) = \min_{C \text{ prefix-free}} L(C).$$

### 2.3 Extensions of Symbol-wise Coding

**Block coding**: Instead of encoding each symbol separately, we can **encode blocks of  $n$  symbols** together. We consider instead the **alphabet  $\mathcal{X}^n$** , and the distribution  $P_{X^n}$ .

This allows us to achieve expected codeword lengths closer to the entropy, as  $n$  increases.  $\forall \epsilon > 0$ , there exists a block code  $C$  s.t.:

$$L(C) < H(X) + \epsilon.$$

#### Advantages of block coding:

- Encode sources with memory, by treating blocks as symbols.  
 $P(\text{apple}) > P(\text{ealpp})$  despite containing the same characters.
- Achieve expected codeword length arbitrarily close to the entropy. Even with independent symbols,

$$NH(X) \leq L(C) \leq NH(X) + 1.$$

$$H(X) \leq \text{Average length per symbol} \leq H(X) + \frac{1}{N}.$$

#### Disadvantages of block coding:

- Determining the product distribution  $P_{X^n}$  can be difficult, especially for sources with memory.
- Encoding and decoding complexity grows exponentially with  $n$ . Requires sorting  $|\mathcal{X}|^n$  symbols by probability, and storing the codebook of size  $|\mathcal{X}|^n$ .